National University of Computer and Emerging Sciences

Technical & Business Writing (SS2012)
Date:24th, September, 2025
Course Instructor(s)
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Sessional-I Exam
Total Time: 1 Hour
Total Marks: 30
Total Questions: **02**

Semester: FL-2025
Campus: Karachi
Dept: CY, SE, AI

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Student Name Roll No Section Student Signature

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Vetted by Vetter Signature

CLO #1: Utilize efficient writing style for producing an effective technical document.

Question 1: Do as directed.

A. Each of the following sentences lack the element of SCOPE given in the bracket. Rewrite these sentences to incorporate the missing element. [1+1+1+2+2 marks]

1. An error is causing the app to crash; the dev team will run tests for this. **(Clarity)**
2. The backend programmers are super lazy and keep delaying API development. **(Objectivity)**
3. The committee agreed and reached a unanimous consensus on the finalized outcome.**(Redundancy)**
4. I would like to inform you that I changed the code lines 45 to 52, for the elimination of the error being caused due to the incorrect lines. Also, according to the discussion we carried out earlier, I have changed the syntax in the first half of the document. **(Economy & nominalizations).**
5. I am just dropping you a few lines to put my name in for that software developer job I saw somewhere in the jobsinpak.com a few weeks back.**(Completeness & formality)**

B. Which statement demonstrates <u>simplicity</u> while explaining a neural network? [1 marks]

a) A neural network is analogous to a multi-layered perceptron that uses stochastic gradient descent with back propagation to minimize cross-entropy loss.
b) A neural network is a computer system inspired by the human brain that learns patterns from data.
c) Neural networks are magic boxes that understand everything automatically.
d) Neural networks are complicated; just use them.

C. Which of the following statements precisely reports a bug to the testing team? [1 marks]

a) The login module crashes when a user enters more than 50 characters in the username field.
b) There's a problem somewhere in the login module.

c) The app is not working properly most of the time.
d) Users are complaining about some random issues in the system.

D. Chose the appropriate purpose, for writing each document, from the given options.[4 marks]

(a) Analyze the causes of the failure of a component, and report your findings so the customer queries can be resolved. **(Persuade, Inform , Instruct)**

(b) Present the blueprint of your project concept. **(Instruct, Persuade, Examine)**

(c) Communicate the steps of a technical task with team. **(Argue, Examine, Instruct)**

(d) A user-guide, compiled from recurring customer queries, to facilitate software installation. **(Persuade, Instruct, Inform)**

E. Match each stage of the technical writing process with its correct description and write only the correct alphabet *(Column B)* with corresponding number *(Column A)* in your answer copy. [5 Marks]

	Column A		Column B
1	Planning	**a**	Adapting the same report into a slide deck for stakeholders
2	Drafting	**b**	Producing a working version with incomplete data visualizations
3	Revising	**c**	Removing redundant sections after peer feedback
4	Editing	**d**	Anticipating whether the audience requires descriptive, procedural, or persuasive form
5	Publishing	**e**	Correcting inconsistent use of units and abbreviations

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Q2. Write an informative paragraph on *"Machine Learning Algorithms"* after synthesizing the given sources (A-C). Use the *three writing strategies* to synthesize and integrate ideas.

Instructions: **[12 marks]**

- **Word limit: Between 70-80 words *(It is mandatory to follow the word limit)***
- **Provide the source reference where necessary.**
- **You must *NOT* add any information of your own.**

Source A: Recent advancements in anomaly detection leverage machine learning models such as Support Vector Machines (SVMs), Random Forests, and Deep Neural Networks. These algorithms analyze network traffic patterns to distinguish between normal and suspicious behaviors. According to a 2024 study, "deep learning models achieve detection rates exceeding 95% when trained on large, labeled datasets."

Source B: Network-level anomaly detection focuses on monitoring real-time traffic flows, packet behavior, and bandwidth utilization. By examining these patterns, cyber-security systems can pinpoint suspicious activities, such as unusual spikes in data transfer or unexpected communication attempts. This approach enables administrators to act quickly, reducing response times to potential attacks.

Source C: Recent research compares supervised and unsupervised learning methods for AI-driven anomaly detection. Supervised techniques, such as decision trees and logistic regression, rely on labeled datasets to classify traffic as normal or malicious, offering high detection accuracy when sufficient labeled data is available. In contrast, unsupervised models, including k-means clustering and auto-encoders, excel at discovering unknown attack patterns in unlabeled datasets. Studies show that a hybrid approach, combining both supervised and unsupervised learning, achieves the highest detection rates, making it ideal for evolving cyber security threats.